

Comparison of Text Representations and Chunk Selection with Abstention for Evidence Retrieval from German Examination Regulations

Seminar Participant / Author

Institut für Sprache und Information
Heinrich-Heine-Universität Düsseldorf
TODO: University Email Address

Abstract

This paper investigates the retrieval of evidence passages in a hierarchically structured German examination regulation. In addition to answerable questions, the task includes an explicit abstention decision for questions that cannot be answered from the source document. The project-specific dataset contains 720 questions, including 120 deliberately unanswerable zero-gold questions, and 201 retrieval chunks. The sparse TF-IDF representation and two fixed multilingual dense representations, Sentence-BERT and multilingual E5, are compared. Building on these, four families of selection methods are evaluated: top- k , absolute threshold, top- k with absolute threshold, and selection via a relative score margin. The hyperparameters are determined exclusively on the development split. Five finalists are subsequently frozen for validation; the winner selected there is evaluated exactly once on the held-out test split. The selected configuration, comprising multilingual E5 and top-1 with threshold ($\theta = 0.83959$), achieves a mean question-wise chunk-selection F1 score of 0.647 on the development split, 0.611 on the validation split, and 0.491 on the test split. The complete review of the 74 test cases that were not exactly correct identifies 21 retrievals of similar but incorrect chunks, 18 threshold-induced abstentions for answerable questions, 13 false-positive retrievals for zero-gold questions, 9 structural top-1 limitations, 9 cases involving valid alternative evidence passages, and 4 problems involving question wording or lexis. An evidence group unseen in the development and validation splits comprises eight test questions and results in $F1 = 0$ in all eight cases. This illustrates how the data split provides a strict test of generalization to unseen evidence groups while preventing evidence leakage.

1 Introduction

Examination regulations combine general procedural rules, subject-specific provisions, and tabular

study-plan information in a hierarchically structured document. Answering a specific question therefore requires more than finding a thematically similar passage. The selected evidence passage must actually contain the rule in question while also being assigned to the correct document level. Passages that use the same technical terms but serve different functions, such as general requirements and their subject-specific implementation, are particularly challenging. Short table cells and entries from study plans can also often be interpreted unambiguously only in conjunction with their headings.

This paper treats the identification of such evidence passages as a separate retrieval task. The prediction target is a set of chunk IDs that provides minimal and complete evidence for a question; the formulation of a natural-language answer is not evaluated. This distinction allows errors in evidence selection to be measured separately from errors made by a downstream reader or generator. Retrieval can serve as an explicit non-parametric knowledge component in knowledge-intensive QA and RAG systems (Lewis et al., 2020); dense bi-encoder methods also implement this separation for open question-answering tasks (Karpukhin et al., 2020). However, providing an evidence passage implies neither that an answer generated later is correct nor that hallucinations are prevented. The investigation is limited to whether the evidence required for an answer is found.

A pure ranking task does not fully represent the application scenario. Plausibly worded questions may presuppose facts that are not regulated in the examination regulation. A system that always outputs at least one chunk inevitably produces an unsupported retrieval in such cases. The task therefore also requires the explicit prediction of the empty set $[\]$. The scientific relevance of such no-answer decisions is established in question answering with deliberately unanswerable questions

(Rajpurkar et al., 2018). In the present retrieval setting, this idea is applied to evidence sets: an abstention is correct exactly when no gold chunk exists for the question.

This problem gives rise to three research questions:

- RQ1** How do sparse lexical and dense semantic representations differ in their ability to retrieve answer-supporting chunks from a German examination regulation?
- RQ2** How do different strategies for chunk selection and abstention affect retrieval performance and abstention behavior?
- RQ3** How well does the selected configuration generalize to the held-out test split, and which observed error patterns and properties of the data split and evaluation are associated with the remaining errors?

To answer these questions, TF-IDF, a multilingual Sentence-BERT checkpoint, and multilingual E5 are compared using the same cosine-similarity measure. Four selector families separate the quality of the representation from the rule that determines which chunks are output or discarded. The dataset comprises 720 questions and 201 chunks from the 2026 examination regulation for the bachelor’s degree program in computational linguistics. Questions with the same non-empty gold evidence set remain in the same group during splitting so that highly similar questions about the same evidence passage are not distributed across the development, validation, and test splits. The paper thus provides a controlled case study of a single German regulatory document: it compares three representations and four selection principles, evaluates an explicit abstention option, and analyzes all test predictions that are not exactly correct.

2 Motivation and Related Work

2.1 Sparse and Dense Text Representations

The classical vector space model represents documents and queries as vectors whose similarity can be determined, among other methods, using cosine similarity (Salton et al., 1975). TF-IDF weights terms based on their occurrence in a text and their specificity within the document collection (Spärck Jones, 1972). For an examination regulation, this principle provides a natural basis for

comparison because questions and evidence passages often use the same technical terms, module names, or regulatory terminology. However, the experiment does not isolate whether this lexical structure actually causes the measured performance; it can serve only as a plausible explanation for the observed TF-IDF results.

Dense representations map texts into continuous vector spaces and can thus capture similarity even when the query and passage do not use the same surface forms. Sentence-BERT produces sentence representations using pair or triplet structures that can be compared directly through cosine similarity (Reimers and Gurevych, 2019). For the multilingual variant, an English-language Sentence-BERT model serves as a reference: a multilingual model is trained on translated sentence pairs so that sentences with the same content receive similar vectors regardless of their language (Reimers and Gurevych, 2020). In the present comparison, Sentence-BERT represents a general multilingual sentence-embedding checkpoint. E5, by contrast, was trained contrastively on text pairs and explicitly developed for tasks involving universal vector representations, including retrieval (Wang et al., 2022). The multilingual E5 family extends this training principle to many languages (Wang et al., 2024). This difference in training objective motivates the comparison but does not in itself provide a causal explanation for a project-specific result.

At the same time, broad benchmarks show why neither sparse nor dense methods should be regarded as fundamentally superior across tasks and corpora. BEIR evaluates zero-shot retrieval across heterogeneous datasets and identifies BM25 as a robust lexical baseline (Thakur et al., 2021). MTEB broadens the perspective to different embedding tasks, datasets, and languages; no single representation method dominates across all tasks (Muenighoff et al., 2023). Rather than adding another general benchmark to these broad comparisons, the present study examines in a controlled manner how three specified representations behave in a small, terminologically narrow corpus.

2.2 Abstention and the Legal Retrieval Context

Selective prediction gives a prediction system the option to reject cases based on a selection criterion. Geifman and El-Yaniv (2019) treat this rejection option as a separate learning problem; the present paper does not train such a model but instead uses a

simple selection rule based on similarity scores. In question answering, SQuAD 2.0 shows that unanswerable but superficially plausible questions require a separate decision (Rajpurkar et al., 2018). Applied to evidence retrieval, this means that the ranking of existing chunks is not the only relevant factor. The system must also recognize when no chunk should be output as evidence.

Work on German legal information retrieval illustrates that matching queries to legal sources is studied as a distinct evaluation problem. GerDaLIR links search queries to relevant German court decisions and compares lexical and neural methods for this purpose (Wrzalik and Krechel, 2021). GerLayQA links legal questions phrased in lay terms and lawyers’ answers to specific statutory sections (Büttner and Habernal, 2024). For large collections of legal documents, Reuter et al. (2025) also investigate how the representation of structurally similar documents affects retrieval reliability in RAG systems. These tasks differ substantially from the examination regulation considered here in terms of corpus size, document type, and target variable. What they have in common, however, is that the quality of the basis for an answer can be assessed separately by linking it to a documented legal or regulatory source. The present paper therefore positions itself as a narrowly defined case study of evidence selection and abstention, not as a general study of legal NLP.

3 Methodology

3.1 Text Representations and Similarity

All representations map the same 720 questions and 201 chunks. For TF-IDF, `TfidfVectorizer(lowercase=True, ngram_range=(1,3))` is used. The vocabulary is learned exclusively from the chunk texts; the questions are then transformed into the same feature space. The configuration thus considers unigrams, bigrams, and trigrams and converts characters to lowercase. Term-frequency scaling is not enabled.

The checkpoint `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` serves as a general dense sentence representation. Its 384-dimensional vectors are generated through the `SentenceTransformer` interface, with no custom pooling method added. The retrieval-oriented alternative is `intfloat/multilingual-e5-base`, which uses 768 dimensions; in accordance with

the intended asymmetric use, questions receive the prefix query: and chunks the prefix passage:. Neither of the two dense models is fine-tuned on the project corpus.

For each representation, cosine similarity

$$s(q, c) = \frac{\mathbf{v}_q^\top \mathbf{v}_c}{\|\mathbf{v}_q\|_2 \|\mathbf{v}_c\|_2} \quad (1)$$

is calculated between every question q and every chunk c , producing a matrix with 720×201 entries for each representation. For the neural encoders, `normalize_embeddings=False` is set; the stored vectors are therefore not L2-normalized in advance, and normalization instead takes place as part of the cosine calculation.

3.2 Selection Rules

The similarity matrix determines only a ranking or score distribution. Four selector families determine which chunks are actually predicted. For a given question, pure top- k returns the k chunks with the highest similarity scores:

$$S_{\text{top-}k}(q) = \arg \text{top}_k \{s(q, c) \mid c \in \mathcal{C}\}. \quad (2)$$

As long as at least k candidates exist, the output is never empty. The selector therefore cannot refuse selection without an additional rule.

An absolute threshold θ , by contrast, selects all chunks whose similarity reaches the cutoff:

$$S_\theta(q) = \{c \in \mathcal{C} \mid s(q, c) \geq \theta\}. \quad (3)$$

This rule can output the empty set, one chunk, or multiple chunks, thereby enabling abstention without limiting the size of the evidence set.

The combined selector first orders all chunks with $s(q, c) \geq \theta$ by descending similarity and returns at most the first k of them, thus combining an absolute minimum similarity with a fixed upper bound on the output size. For $k = 1$, the result is either exactly one evidence passage or `[]`.

The relative margin evaluates the separation at the decision boundary rather than the absolute score level. With s_k denoting the similarity score at rank k and s_{k+1} the following score,

$$m_k(q) = \frac{s_k - s_{k+1}}{\max(|s_k|, 10^{-12})}. \quad (4)$$

The hyperparameter γ specifies the minimum required size of this relative distance. If $m_k(q) \geq \gamma$, the first k chunks are output; otherwise, the prediction is `[]`. For $k = 1$, the rule measures how clearly

the best chunk is separated from the second-best candidate in relative terms. A high absolute peak score is neither necessary nor sufficient for this criterion.

3.3 Determination of the Search Ranges

Because the three representations produce different similarity distributions, no shared threshold ranges are used. The initial regions are derived from robust quantiles of three score distributions on the development split: the respective lowest similarity scores of required gold chunks, the highest scores of non-gold chunks, and the highest scores for zero-gold questions. This procedure restricts the search to regions in which correct retrieval and abstention can be distinguished without using validation or test data. The documented initial ranges are 0.8331 to 0.8865 for E5, 0.0591 to 0.5162 for TF-IDF, and 0.4740 to 0.7595 for Sentence-BERT. All evaluations used to select k , θ , and γ are conducted exclusively on the development split.

4 Experimental Setup

4.1 Corpus, Chunking, and Gold Standard

The corpus is based on the 2026 examination regulation for the bachelor’s degree program in computational linguistics. The active retrieval collection contains 201 chunks totaling 10 194 words, with an average of 50.72 words and a median of 41 words per chunk; the range is 8 to 204 words. In terms of content, 133 chunks belong to the general examination regulation, 30 to the subject-specific appendix, 36 to the exemplary study plan, and 2 to the front matter. Short excerpts from tables or appendices include their hierarchical context headings so that their meaning remains recognizable outside the original page.

The question dataset comprises 720 entries. Of these, 600 can be answered on the basis of the corpus, while 120 zero-gold questions deliberately have no supporting passage. For each answerable question, `all_required_chunk_ids` denotes a minimally intended, complete evidence set. This target thus contains the chunks jointly required for the intended answer but does not claim to capture every semantically equivalent alternative evidence passage in the document. Table 1 shows the distribution of empty, single-part, and two-part gold sets.

TODO: document which steps in chunking, question creation, gold assignment, and quality control

Split	Total	0 Gold	1 Gold	2 Gold
Development	432	74	340	18
Validation	144	24	116	4
Test	144	22	112	10
Total	720	120	568	32

Table 1: Distribution of gold evidence sets across the three data partitions.

were performed manually, with script support, or with language-model assistance.

4.2 Data Split and Model Selection

The data are divided into 432 development, 144 validation, and 144 test questions. The governing grouping rule is `same_non_empty_gold_chunk_set`: all questions with the same non-empty gold evidence set remain in the same data partition. This prevents an evidence passage tested through several paraphrased questions in the test split from having already appeared in a development or validation question with the same gold set. The split contains 315 groups, the largest of which comprises 13 questions, and no non-empty gold set occurs in more than one split. Zero-gold questions each form their own group because they would otherwise be combined into a single large group solely on the basis of their identical empty sets.

This grouping reduces evidence leakage but makes the generalization criterion stricter. The test system must not only process new question formulations but also sometimes retrieve evidence passages whose gold set did not occur as a target in either development or validation. This distinction is central to interpreting the test score: a decline in performance may be associated with the stricter separation of evidence groups. Without comparison to a split without evidence grouping, however, its influence cannot be quantified.

Selector families and hyperparameters are compared exclusively on the development split. The primary ranking criterion is the mean question-wise F1 score; ties are broken by exact match, mean question-wise precision, and, finally, a lower average number of selected chunks. Five finalists are frozen before validation. Only these configurations are compared on the validation split; no readjustment based on validation takes place. The winner selected there is then run unchanged exactly once on the test split.

4.3 Metrics and Comparison Methods

For each question, the predicted set P_q and gold set G_q are compared. If both sets are non-empty, precision and recall are given by $|P_q \cap G_q|/|P_q|$ and $|P_q \cap G_q|/|G_q|$, respectively; their harmonic mean is the question-wise F1 score. For $P_q = G_q = []$, precision, recall, and F1 are assigned a value of 1 because the system abstains correctly. If exactly one of the two sets is empty, F1 is instead 0. The primary metric, Q-F1, is the arithmetic mean of these question-wise scores. It thus weights each question equally, regardless of the size of its gold set.

A completely missing prediction record must be semantically distinguished from an explicit empty prediction. Missing entries are excluded from the means and counted separately as unanswered. For the frozen test evaluation, prediction records are available for all 144 questions. In addition, mean question-wise precision and recall, exact match (EM), micro-F1, zero-gold abstention rate, rate of empty selections, and average number of selected chunks are reported.

Three simple comparison methods contextualize the test performance. *Random top-1* selects one uniformly random chunk per question; the mean and standard deviation across the fixed seeds 0 to 99 are reported. The *most frequent development answer* predicts the most frequent non-empty exact gold set in the development split for every validation and test question. *Always no selection* returns $[]$ regardless of the question. This final method directly measures the Q-F1 attainable solely from the proportion of zero-gold questions.

The experiments were conducted using Python 3.11. Packages central to reproduction include `scikit-learn==1.9.0`, `sentence-transformers==3.4.1`, `transformers==4.46.3`, `torch==2.1.2` as a CPU build, and `numpy==1.26.4`. The complete frozen dependency list is documented in the project artifact `requirements.txt`.

5 Results and Evaluation

5.1 Development Results

Table 2 reports the best configuration found on the development split for each combination of representation and selector family, with E5 consistently achieving the highest Q-F1 scores among the evaluated representations. The best E5 configuration combines top-1 with the absolute threshold $\theta =$

0.83959 and achieves Q-F1 = 0.6474. The difference relative to pure top-1, with Q-F1 = 0.6165, is 0.0309; relative to the best pure threshold selector, with Q-F1 = 0.4941, it is 0.1533. The relative margin lies between these configurations, with Q-F1 = 0.6312.

TF-IDF remains competitive despite the absence of dense semantics: its best configuration is the relative margin, with Q-F1 = 0.5702, narrowly ahead of top-1 with threshold at 0.5671. Thus, the ordering of the selectors is not consistent across representations. The evaluated Sentence-BERT checkpoint achieves its best Q-F1 of 0.3989 with top-1 and threshold and ranks behind E5 and TF-IDF throughout the experimental setup. This result applies only to the specific non-fine-tuned checkpoint and the present dataset.

The differences between the selectors show the trade-off between retrieval and abstention. Pure top-1 effectively limits the output but cannot abstain on any question. Pure threshold selection enables empty and multi-part outputs but exhibits a marked gap between precision and recall for all three representations. For E5, for example, a mean precision of 0.4578 contrasts with a recall of 0.6285. Top-1 with threshold combines the abstention option with an upper bound of one chunk; the relative margin, by contrast, uses the local separation between the two highest scores. In the development results, which rule is more favorable depends on the representation.

5.2 Validation and Held-Out Test

Among the five finalists frozen in advance, E5 with top-1 and $\theta = 0.83959$ achieves the highest Q-F1 on the validation split and is therefore selected as the final system. Sentence-BERT is not among these five finalists; validation therefore does not constitute another complete comparison of all three representations. As Table 3 shows, the selected configuration achieves Q-F1 = 0.6111, EM = 0.5972, and a zero-gold abstention rate of 0.4167 on validation.

On the held-out test split, Q-F1 decreases to 0.4907, an absolute decline of 0.1204 relative to validation; EM is 0.4861, and micro-F1 is 0.5041. The configuration outputs an average of 0.7917 chunks and returns the empty set for 20.83% of questions. Of the 22 zero-gold questions, 9 are correctly rejected, corresponding to an abstention rate of 0.4091. However, the remaining errors cannot be explained by a one-sided tendency to produce

Representation	Selector	Parameter	Q-F1	EM	Precision	Recall
E5	Top-1 + Threshold	$\theta = 0.83959$	0.6474	0.6366	0.6528	0.6447
E5	Relative Margin	$\gamma = 0.0025$	0.6312	0.6204	0.6366	0.6285
E5	Top-1	$k = 1$	0.6165	0.6042	0.6227	0.6134
E5	Threshold	$\theta = 0.866475$	0.4941	0.3657	0.4578	0.6285
TF-IDF	Relative Margin	$\gamma = 0.235$	0.5702	0.5579	0.5764	0.5671
TF-IDF	Top-1 + Threshold	$\theta = 0.0999$	0.5671	0.5532	0.5741	0.5637
TF-IDF	Top-1	$k = 1$	0.5509	0.5370	0.5579	0.5475
TF-IDF	Threshold	$\theta = 0.127665$	0.4866	0.3657	0.4596	0.5706
Sentence-BERT	Top-1 + Threshold	$\theta = 0.6751$	0.3989	0.3958	0.4005	0.3981
Sentence-BERT	Relative Margin	$\gamma = 0.005$	0.3866	0.3819	0.3889	0.3854
Sentence-BERT	Top-1	$k = 1$	0.3742	0.3681	0.3773	0.3727
Sentence-BERT	Threshold	$\theta = 0.7095375$	0.3583	0.2801	0.3343	0.4421

Table 2: Best development configuration for each representation and selector family. Precision and recall are mean question-wise metrics.

either too many or too few outputs: in addition to false-positive retrievals for zero-gold questions, empty predictions occur for answerable questions. Because no repeated splits or comparable uncertainty analysis are available, the difference between validation and test is not interpreted as statistically expected variation.

The comparison methods in Table 4 show that the test score is not achieved by trivial strategies. Random top-1 achieves a mean Q-F1 of only 0.0047 across 100 seeds. The most frequent non-empty development answer matches no complete test gold set and therefore achieves Q-F1 = 0. By contrast, always no selection achieves 0.1528 because this method handles exactly the 22 zero-gold questions correctly and all 122 answerable questions incorrectly. The final system clearly exceeds these reference points, although its absolute test performance remains limited.

5.3 Error Analysis and Contextualization

The manual review covers all 74 test questions without an exact match. Table 5 assigns a primary error category to each case. The largest group, comprising 21 cases, involves the retrieval of a similar chunk that is nevertheless incorrect for the specific question. Such confusions correspond to the core problem of the corpus: lexically or thematically similar passages can serve different regulatory functions. A further 18 cases arise because the best correct candidate does not exceed the absolute threshold, causing the system to abstain on an answerable question. Conversely, it retrieves a chunk for 13 unanswerable questions. The similar sizes of these two groups suggest that the errors cannot be fully resolved merely by shifting the threshold

globally.

A second source of error is the fixed top-1 restriction. According to the gold standard, ten test questions require two chunks, so a selector that can output at most one chunk can never achieve an exact match in these cases. If exactly one of the two required chunks is retrieved correctly, this yields $P = 1$, $R = 0.5$, and thus at most

$$F1 = \frac{2 \cdot 1 \cdot 0.5}{1 + 0.5} = \frac{2}{3}. \quad (5)$$

Nine cases that are not exactly correct are assigned primarily to this structural limitation in the audit. The finding thus concerns not only the quality of the E5 representation or the position of the threshold but also the predetermined maximum output size.

The effect of the group-based data split is particularly visible in evidence group G0040. It comprises the eight test questions Q0153 to Q0160 with the shared gold chunk P025CL-GEN-C03-P12-A02; this non-empty gold set occurs in neither development nor validation. All eight questions achieve F1 = 0. Q0153, Q0154, Q0155, Q0156, Q0158, and Q0159 each return an incorrect non-empty chunk, while Q0157 and Q0160 abstain with []. The cluster is therefore a verified example of failure to generalize to an unseen evidence group. The fact that this evidence group remains entirely unseen is consistent with the deliberately stricter split criterion and does not indicate that the test split was constructed incorrectly.

At the same time, the evaluation is limited by the exhaustiveness of the gold standard. Following the completed review, nine formally incorrect chunk IDs contain a semantically complete alternative evidence passage. These cases are classified as

Split	Q-F1	EM	Precision	Recall	Micro-F1	ZG Abst.
Validation	0.6111	0.5972	0.6181	0.6076	0.6475	0.4167
Test	0.4907	0.4861	0.4931	0.4896	0.5041	0.4091

Table 3: Frozen winner E5 top-1 with $\theta = 0.83959$ on validation and test. For the test, 144 out of 144 predictions are available; TP = 62, FP = 52, and FN = 70.

System	Q-F1	Precision	Recall	EM	ZG Abst.
Random Top-1	0.0047 ± 0.0049	0.0050 ± 0.0050	0.0045 ± 0.0048	0.0041 ± 0.0048	0.0000
Most Frequent Development Answer	0.0000	0.0000	0.0000	0.0000	0.0000
Always No Selection	0.1528	0.1528	0.1528	0.1528	1.0000
E5 Top-1 + Threshold	0.4907	0.4931	0.4896	0.4861	0.4091

Table 4: Test performance of the final system and the frozen comparison methods. For random top-1, the mean and standard deviation across 100 fixed seeds are reported.

Error Category	n	Share
Similar, Incorrect Chunk	21	28.4%
False Abstention	18	24.3%
False Positive for Zero-Gold	13	17.6%
Top-1 Limitation	9	12.2%
Valid Alternative Evidence Passage	9	12.2%
Question Wording / Lexis	4	5.4%
Total	74	100.0%

Table 5: Primary categorization of all test cases that are not exactly correct.

valid_alternative_evidence but remain incorrect in the official evaluation because they do not belong to the minimally intended gold set. The official Q-F1 of 0.4907 is therefore not subsequently replaced. A purely post-hoc sensitivity calculation can illustrate the magnitude of this effect together with that of the unseen group G0040: if the nine alternative evidence passages are counted as fully correct and the eight G0040 cases with F1 = 0 are excluded from the diagnosis, the result is

$$\frac{70.6667 + 9}{144 - 8} \approx 0.5858. \quad (6)$$

This value is neither a corrected test result nor a new model evaluation. It merely describes the extent to which two previously identified properties of the dataset and evaluation can influence the raw gap.

For RQ1, the development results thus show a clear project-specific ordering: E5 ranks ahead of TF-IDF, while the evaluated Sentence-BERT checkpoint performs less well. The retrieval-oriented contrastive pretraining of E5 is compatible with this finding but is not tested as an isolated cause. Likewise, the terminological specificity of the examination regulation is a plausible explanation for the

relatively strong TF-IDF performance, although the experiment does not provide corresponding causal evidence. For RQ2, there is no universally best selector. Top-1 with absolute threshold is the best E5 configuration and wins on validation; for TF-IDF, the relative margin is slightly ahead on development. The test answers RQ3 by showing weaker generalization than on validation and several types of error operating simultaneously. In addition to ranking and threshold errors, the fixed top-1 limit, unseen evidence groups, and non-exhaustive gold alternatives contribute to the measured result.

6 Conclusion

The investigation deliberately separates the selection of evidence passages from subsequent answer generation and compares three text representations under the same retrieval and evaluation conditions. With regard to RQ1, multilingual E5 achieves the highest scores among all evaluated representations on the development split; the configuration selected from it also ranks first among the five finalists on validation. Nevertheless, TF-IDF achieves substantially higher scores in the small, terminologically specific corpus than the investigated multilingual Sentence-BERT checkpoint. This ordering applies to the present dataset and the checkpoints used, not to other legal or administrative corpora.

For RQ2, the selection rule proves to be a distinct component of the system. Pure top-1 already achieves strong development results but has no abstention option, whereas a pure threshold can abstain but does not limit the number of output chunks. The combination of E5, top-1, and $\theta = 0.83959$ selected on validation combines ranking, an upper bound of one chunk, and the possi-

bility of an empty output. The fact that the relative margin performs slightly better for TF-IDF instead argues against the general dominance of this selector.

RQ3 shows the limitations of the selected configuration. Q-F1 decreases from 0.6111 on validation to 0.4907 on the held-out test. The complete review of all 74 cases that are not exactly correct does not attribute this decline to a single cause. Incorrect similar chunks, false abstentions, retrievals for zero-gold questions, the top-1 restriction, alternative valid evidence, and problems with individual question wording occur alongside one another. The eight entirely incorrect questions in the unseen evidence group G0040 illustrate the stricter generalization resulting from the separation of identical gold sets. Evidence retrieval thus remains a clearly measurable upstream component for subsequent QA or RAG systems; the quality of generated answers cannot be inferred from the present results.

7 Limitations and Future Work

The empirical scope is limited to one regulatory document for one degree program. Transferability to other examination regulations or universities, as well as to larger collections of legal documents, was not investigated. The questions and gold evidence sets were also constructed specifically for the project by the author. Because no independent second annotation was conducted, inter-annotator agreement could not be assessed. Before submission, the provenance of chunking, question creation, assignment, and review must also be reported in full; without this information, the proportions of human, script-supported, and language-model-assisted work cannot be described reliably.

The evaluation target represents a minimally intended complete evidence set but not an exhaustive set of all permissible evidence passages. The nine test predictions reviewed as valid alternatives show the practical consequence of this decision. They do not justify a subsequent change to the official test score, but they support future gold standards that explicitly model equivalent evidence sets or canonicalized references. In addition, the final top-1 selector limits the attainable performance for questions with multiple required evidence passages; this concerns ten questions in the test split alone.

Abstention is also not conclusively calibrated. The system identifies 9 of 22 zero-gold questions but incorrectly retrieves a chunk for the remain-

ing 13. At the same time, 18 threshold-induced abstentions occur for answerable questions. Characterizing the system merely as too conservative or too permissive therefore falls short. Future experiments should investigate calibration separately by question type and permit selectors that output zero, one, or multiple chunks depending on the question.

The frozen comparison includes neither BM25 nor hybrid sparse-dense methods, cross-encoder reranking, or generative reader models. BEIR identifies BM25 as a robust zero-shot baseline (Thakur et al., 2021); however, a direct comparison on the present corpus remains outstanding. Meaningful extensions therefore include BM25 and hybrid methods, downstream reranking of similarly scored chunks, adaptive evidence sets, and content-type-aware strategies for running text, appendices, and tables. Downstream answer generation should be evaluated only after such an extension of retrieval. RAG provides a suitable system framework for this purpose (Lewis et al., 2020), but answer correctness and faithfulness to evidence would then have to be measured in addition to, and separately from, the retrieval performance investigated here.

References

- Marius Büttner and Ivan Habernal. 2024. [Answering legal questions from laymen in German civil law system](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2015–2027. Association for Computational Linguistics.
- Yonatan Geifman and Ran El-Yaniv. 2019. [SelectiveNet: A deep neural network with an integrated reject option](#). In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. [Dense passage retrieval for open-domain question answering](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781. Association for Computational Linguistics.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.

- Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. 2023. [MTEB: Massive text embedding benchmark](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2014–2037. Association for Computational Linguistics.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. [Know what you don’t know: Unanswerable questions for SQuAD](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789. Association for Computational Linguistics.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-BERT: Sentence embeddings using siamese BERT-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992. Association for Computational Linguistics.
- Nils Reimers and Iryna Gurevych. 2020. [Making monolingual sentence embeddings multilingual using knowledge distillation](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4512–4525. Association for Computational Linguistics.
- Markus Reuter, Tobias Lingenberg, Ruta Liepina, Francesca Lagioia, Marco Lippi, Giovanni Sartor, Andrea Passerini, and Burcu Sayin. 2025. [Towards reliable retrieval in RAG systems for large legal datasets](#). In *Proceedings of the Natural Legal Language Processing Workshop 2025*, pages 17–30. Association for Computational Linguistics.
- Gerard Salton, Anita Wong, and Chung-Shu Yang. 1975. [A vector space model for automatic indexing](#). *Communications of the ACM*, 18(11):613–620.
- Karen Spärck Jones. 1972. [A statistical interpretation of term specificity and its application in retrieval](#). *Journal of Documentation*, 28(1):11–21.
- Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. 2022. [Text embeddings by weakly-supervised contrastive pre-training](#). *arXiv preprint arXiv:2212.03533*.
- Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. [Multilingual E5 text embeddings: A technical report](#). Technical Report MSR-TR-2024-45, Microsoft Research.
- Marco Wrzalik and Dirk Krechel. 2021. [GerDaLIR: A German dataset for legal information retrieval](#). In *Proceedings of the Natural Legal Language Processing Workshop 2021*, pages 123–128. Association for Computational Linguistics.